

# Topic 5: Arms Race

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# Overview



**Scenario:** Purple (P) and Green (G) are two countries competing in an arms race.

**Research Question:** What happens to the long-term spending of each country?

## Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- How does public sentiment affect defence spending?
- How does limiting spending affect a country's position?

**Motivation:** It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.



# Base Model

## Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

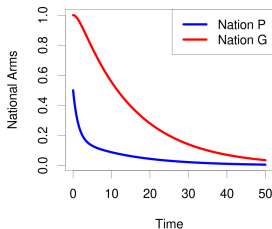
- Feasibility Condition:  $(p(t), g(t) \geq 0)$
- $a, c$ : Mutual Fear ( $a, c > 0$ )
- $b, d$ : Expenditure ( $b, d > 0$ )

# Long Term Behavior (Base Model)



$$bd > ac$$

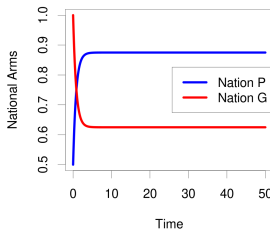
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual  
disarmament

$$bd = ac$$

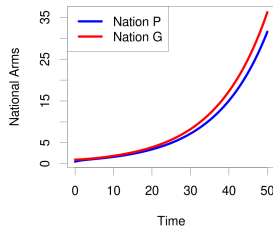
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Perpetual arms race

# Public Sentiment (Trust and Hostility)



## Model Equations

The equations after including public sentiment are:

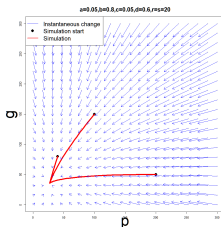
$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

■  $(r, s) > 0$  for trust and  $(r, s) < 0$  for hostility

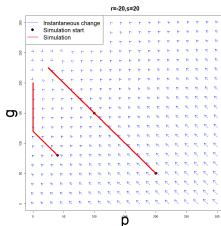
# Long Term Behavior (Public Sentiment)

$$bd > ac \text{ or } bd = ac, r < 0, s < 0$$



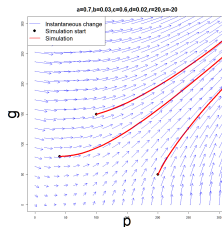
Complete mutual disarmament

$$bd = ac, r < 0, s > 0$$



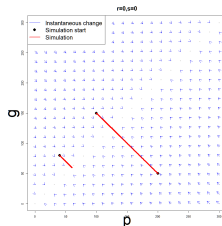
One nation grows, other nation shrinks

$$bd < ac \text{ or } bd = ac, r \geq 0, s > 0$$



Perpetual arms race

$$bd = ac, r = 0, s = 0$$



Nations never reach equilibrium



# Mutual Logistic Fear

**Limitation of the Current Model:** Each country has a limited amount of resources at a given time.

**Solution:** Introduce logistic growth.

## Model Equations

The equations after adding logistic growth to account of limited resources are:

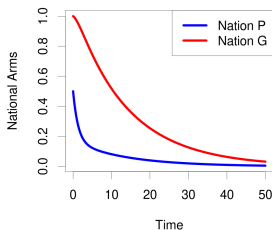
$$\begin{aligned}\frac{\partial p}{\partial t} &= a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r \\ \frac{\partial g}{\partial t} &= c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s\end{aligned}$$

# Long Term Behavior (Mutual Logistic Fear)



$$bd > ac$$

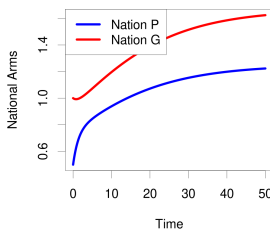
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual  
disarmament

$$bd = ac$$

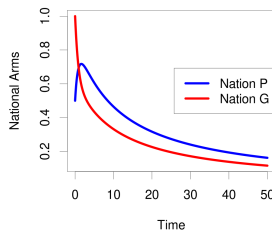
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race



# Thank you for listening!



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